LCA calculations using matrices: summary

1. Introduction to LCA

- LCA is used to **asses** all types of products and products systems.
- It's important to include in the LCA all the various life-cycle **phases** (construction, operation = maintenance, demolition) because their importance varies.
- LCA represents the importance of having a **holistic understanding** of the environmental aspects associated with a service or product.
- The objective is to be able to describe the **total environmental load** associated with the delivery of a **functional unit**. The steps for doing this are:
    - 1. Identify the total activity generated in all processes involved
    - 2. Identify the emissions in each process per unit produced
    - 3. Multiply the emissions factors · activity (step 1 · 2) = emission generated in each node as a result of the requirements of the functional unit, distinguishing direct, indirect and total emissions.
- **Functional unit**: the product / service we are assessin; **Reference flow**: how much products do we need to deliver that product / service? So we scale the amount of products needed. **System boundaries**: What parts of the economy are involved in delivering that reference flow? So we expand to all processes needed ➔ e.g. **FU**: "Drive 1000km with an electric car"; **RF**: 0.005 cars needed (assuming 200k lifetime) + 200 kWh electricity. **SB**: Production and final disposal of 0.005 electric vehicles + electricity generation and distribution = all life cycle.
- Thus, the "final" objective is normally to **compare** consistently technological systems with respect their environmental **impacts,** which involves resolving **system border** issues.
- The challenge is to avoid **making decisions** on development, production and use of technologies based on inconsistent environmental assessments (or even without them).
- It's an important tool to reveal issues of **problem shifting,** and thus asses to avoid them. Problem shifting can be understood like moving the environmental impacts of one product in one phase to another phase (ex: electric vehicles and electricity source production) and also in the meaning of generating another problem at the time of solving one issue. This is specially important currently with the monotone focus in the global warming only.
- When we initiate an **activity** (functional unit), we generate:
    - **Direct** activity / emissions = Activity / emissions generated in the processes which require the functional unit from.
    - **Indirect** activity / emissions = Activity / emissions generated of all other processes as a result of the functional unit = chains of requirements.
    - **Total** activity / emissions = direct + indirect activities / emissions.

2. Essentials LCA calculations

BASIC MATHEMATICS OF LCA

- **A, requirement matrix** = composed by **coefficients of requirements** (a_{ij}) = denotes the amount required by process i per unit of output of process j **= input / output.**
    - **Columns** are requirements for each process per **unit** production = "list of ingredients" of each process (these data requirements needs to be collected.
    - **Rows** are what each process gives per unit process j.
        - It's important to **normalize** the input data by dividing by the total output, and thus obtaining amounts per unit of production.
        - The coefficients like a_{11} (or when i=j) normally are 0, but the relations has to be expressed in a general way. One example could be a bulb factory, that needs also bulbs to illuminate the factory itself.
    - A has to **satisfy** the **Hawkins - Simon condition** = **the (A-I) determinant** has to be:
        - Different from 0 (in order to be invertible)
        - Positive (LCA req. application: all the processes have to be self sustaining).
    - The subparts of the matrix are: (summarizing: flows from **i** to processes **j** ➔ Aij)

$$A = \begin{bmatrix} A_{ff} & A_{fb} \\ A_{bf} & A_{bb} \end{bmatrix}$$

-**Aff** = Relationship of the needs between the foreground processes studied.

-**Afb** = Assumed 0, no processes are going from f. to b.

-**Abf** = Relation / flows / upstream inputs going from the background to the foreground processes / system.

-**Abb** = Flows between background processes.

    - It is generally **convenient** to **distinguish** between the system that you model and investigate in detail (foreground system) and generic data that you use to complete value chains upstream in your system (background system).
    - When the flowcharts are **unidirectional** ➔ processes can be sorted so all input coefficients are in one side of the diagonal.

- **y, external demand** of products (y_i) = **sets the external demand** to one functional unit = **list of ingredients** of the **studied process** (normally from the foreground) needed for 1 functional unit. So y is what you are asking for.
- **x, output vector** = describes the production output of each node **= internal demand** (demand of products between the network processes) **+ external demand** (the requirements of products of the network, normally the functional unit). So it's the supply chain response for a given demand (e.g: 1 kg of steel will need inputs of electricity, transport, etc). x is what the system does to satisfy y (what you are asking for).

- The production, thus is = output.
- To solve the equation above (x = Ax + y), we have to rearrange yields:

$$\overbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}^{\text{output}} = \overbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}^{\text{intermediate demand}} + \overbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}^{\text{external demand}}$$

$$x = Ax + y \quad \Leftrightarrow \quad (I - A)x = y \quad \Leftrightarrow \quad x = (I - A)^{-1}y$$
$$L = (I - A)^{-1} \Rightarrow \quad x = Ly$$

- **L, Leontief inverse matrix** = composed by **coefficients** (L_{ij}) = represents the amount of output of processes **i** that is required per unit of external demand of product from processes **j.**
  - **Columns =** each column represents the vector of the total output required per unit final demand of a given process (= product from the process j).
  - **x vector =** contains the output required for a specific external demand (y)

BASIC CONTRIBUTION ANALYSIS

- **S, stressors matrix / biosphere matrix** = contains the environmental stressors (elementary flows) associated with the output of each process. This data has to be collected analogously to the requirements coefficients of A. These are the emissions per unit of industrial output.
  - Columns = vector stressors for one unit of output of that particular process.
  - The S matrix is **constructed** as follows:

$$S = \begin{bmatrix} S_f & S_b \end{bmatrix}$$

-**Sf** = stressors form the foreground processes

-**Sb** = stressors from the background processes

- **e, stressors vector** = contains **stressors (i.e., elementary / biosphere flows) associated** with a given final / external **demand** (given by the y vector), as an inventory flows vector:

$$e = \begin{bmatrix} e_1 \\ \vdots \\ e_{str} \end{bmatrix} = \begin{bmatrix} s_{11} & \cdots & s_{1,pro} \\ \vdots & \ddots & \vdots \\ s_{str,1} & \cdots & s_{str,pro} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_{pro} \end{bmatrix}$$

  - That is the same as: $e = Sx = SLy = S(I - A)^{-1}y$

- **X^, x hat diagonal matrix** = it means place the x vector on the diagonal of a matrix, with zeros elsewhere, in order to study **how** the **processes contribute** to our given demand. Thus, it serves to later calculate LCA contribution from each process.

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \hat{x} = \begin{bmatrix} x_1 & 0 & 0 \\ 0 & x_2 & 0 \\ 0 & 0 & x_3 \end{bmatrix}$$

- **E, stressors matrix** = inventory matrix of all needed biosphere flows from a given demand. It has the same meaning as the **stressors vector e**, but it explains the contribution of each stressor to the total load of stressors from a given demand:

$$E = \begin{bmatrix} e_{11} & \cdots & e_{1,pro} \\ \vdots & \ddots & \vdots \\ e_{str,1} & \cdots & e_{str,pro} \end{bmatrix} = \begin{bmatrix} s_{11} & \cdots & s_{1,pro} \\ \vdots & \ddots & \vdots \\ s_{str,1} & \cdots & s_{str,pro} \end{bmatrix} \begin{bmatrix} x_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & x_{pro} \end{bmatrix}$$

$$E = S\hat{x} = S\widehat{Ly} = S\widehat{(I-A)^{-1}y}$$

  - **Columns** = express for each stressor (CO_2, etc), the **amount contributed** from each processes.
  - As it is a "detailed" version of vector e, it satisfies that:

$$e = \sum_{pro} E_{str,pro}$$

- **C, characterization matrix** = describes the **relationship between stressors** (identified in the columns) and the **impact categories** (identified by the rows), in order to translate or understand in a better way the results of the stressors = **characteritzation factors midpoints**

$$C = \begin{bmatrix} c_{11} & \cdots & c_{1,str} \\ \vdots & \ddots & \vdots \\ c_{imp,1} & \cdots & c_{imp,str} \end{bmatrix}$$

- **d, vector of impacts** = with a chosen set of impact categories to be evaluated (in the C matrix), we can now calculate the vector of **total impacts** (d), for a given external demand:

$$d = \begin{bmatrix} d_1 \\ \vdots \\ d_{imp} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1,str} \\ \vdots & \ddots & \vdots \\ c_{imp,1} & \cdots & c_{imp,str} \end{bmatrix} \begin{bmatrix} e_1 \\ \ddots \\ e_{str} \end{bmatrix}$$

$$d = Ce = CSx = CSLy = CS(I-A)^{-1}y$$

- **Dpro, matrix of impacts by processes** = shows us how the different processes contribute to the impact categories.

$$D_{pro} = \begin{bmatrix} d_{11} & \cdots & d_{1,pro} \\ \vdots & \ddots & \vdots \\ d_{imp,1} & \cdots & d_{imp,pro} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1,str} \\ \vdots & \ddots & \vdots \\ c_{imp,1} & \cdots & c_{imp,str} \end{bmatrix} \begin{bmatrix} e_{11} & \cdots & e_{1,pro} \\ \vdots & \ddots & \vdots \\ e_{str,1} & \cdots & e_{str,pro} \end{bmatrix}$$

$$D_{pro} = CE = CS\hat{x} = CS\widehat{Ly} = CS\widehat{(I-A)^{-1}y}$$

- **Dstr, matrix of impacts by stressors** = shows us how the different stressors contribute to the impact categories.

$$D_{str} = \begin{bmatrix} d_{11} & \cdots & d_{1,pro} \\ \vdots & \ddots & \vdots \\ d_{imp,1} & \cdots & d_{imp,pro} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1,str} \\ \vdots & \ddots & \vdots \\ c_{imp,1} & \cdots & c_{imp,str} \end{bmatrix} \begin{bmatrix} e_1 & 0 & 0 \\ 0 & \ddots & \vdots \\ 0 & 0 & e_{str} \end{bmatrix}$$

$$D_{str} = C\hat{e} = C\widehat{Sx} = C\widehat{SLy} = C\widehat{S(I-A)^{-1}y}$$

- Both Dpro and Dstr, as it happens with E and e matrices, as they represent the same with the difference that each of them details the information from d, we can **obtain** the d vector by **simply summing** the **rows** of both matrices:

$$d = \sum_{pro} D_{pro} = \sum_{str} D_{str}$$

SUMMARY MATRICES AND VARIABLES

Matrix / variables	Dimensions	Description
A	$pro \times pro$	matrix of inter-process requirements
y	$pro \times 1$	vector of external demand of processes, reference flow
x	$pro \times 1$	vector of outputs for a given external demand
L	$pro \times pro$	matrix of outputs per unit of external demand (Leontief inverse)
S	$str \times pro$	matrix of stressors intensities per unit output (biosphere matrix)
e	$str \times 1$	vector of stressors generated for a given external demand
E	$str \times pro$	matrix of stressors generated from each process for a given external demand
C	$imp \times str$	characterization matrix
d	$imp \times 1$	vector of impact scores for a given external demand
$Dpro$	$imp \times pro$	matrix of impact scores from each process for a given external demand
$Dstr$	$imp \times str$	matrix of impact scores from each stressor for a given external demand

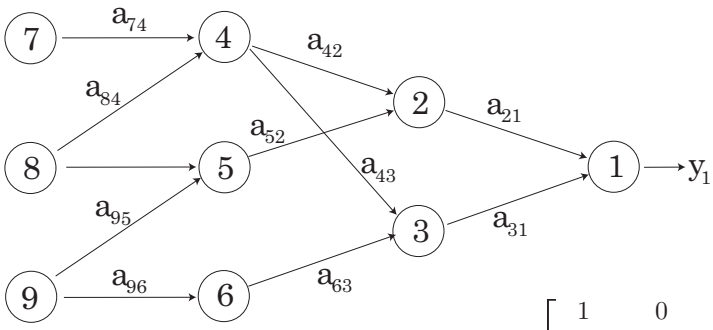
pro	Processes, technosphere flows
str	Stressors, elementary / biosphere flows
imp	Impact categories

➔ It's important to remember the **matrix multiplication rules** (in terms of the range of the matrices) to deduce the **dimensions** of the studied matrices.

- That means ➔ for A = B · C, remember that the **columns** of the **matrix B** have to be the **same** dimensions than the **rows for the C matrix.**
- Also, the dimensions of A will be then (c of B, r of C). This will also help to interpret the meaning of the matrices.

3. Other aspects related with LCA calculations

- **Flowcharts** gives and overview of how different components interact, where the requirements from one node into the production of another node is indicated by arrows.
    - They are normally unidirectional when you look into detail, but zooming out normally are parts of a bigger loop activities that intervene in all sorts and manners.
    - To construct the A matrix (for a unidirectional flow), we simply **numerate** the processes begging from the left and from the upper part (can be also from the bottom). Numerating the arrows with the numeration (ij) we can easily place each value in the corresponding column and row = coefficient a_{ij}:



$$(I - A_{ij}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -a_{21} & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -a_{31} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -a_{42} & -a_{43} & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -a_{52} & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -a_{63} & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -a_{74} & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -a_{84} & -a_{85} & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -a_{95} & -a_{96} & 0 & 0 & 1 \end{bmatrix}$$

- The **geometric series expansion** allow us to understand what **degree** the **impacts** occur, beginning from our external demand and going each time more far to the upstream processes.
    - That means that if we have a unidirectional flow of processes (like 1 ➔ 2 ➔ 3), the first tier (first order of the taylor's expansion) will correspond to the impacts generated by the process 1, while the second tier will correspond to the process 2 and so on.
    - When we proceed this calculations into the background system we will then experience loops, because all the processes in the background will require output from each other = there will be activity and all tiers.
    - Tiers will reflect **rounds of transactions,** and the level of output of these transactions (of the upstream processes) will decrease the more rounds (or tiers) we go.
    - Looking at the accumulated impacts in a plot, we can have a general idea of how the environmental loads are distributed in the different tiers.
    - As the number of tiers goes to the **infinity** ➔ output goes 0 and the series **converge**. The calculation can be done by:

$$x_t = A^t y = A(A^{t-1}y) = Ax_{t-1}$$

- To **include the various life cycle phases in a model**, we define coefficients in the unidirectional value chain that represented the construction and demolition required per unit of process i.
    - This means that coefficients express how much **requirements**, as well as emissions and impacts (associated with the construction and demolition phases) can be **attributed** to per unit product **output in operation phase** = which is the **environmental load associated.**
    - The problem is an analogy of the capital costs that can be attributed to a product, but in this case there is not a discount rate (it is assumed 0) = in life cycle assessment it is well established **not to discount the environmental loads.**
    - Thus, these loads can be calculated with a **linear distribution** to each unit output, by dividing the total load (of construction or demolition phases) by the **total units** produced through the lifetime of the factory:

$$a_{CiOi} = a_{DiOi} = \frac{1}{\text{total production over lifetime}} = \frac{1}{\dot{m}_{year}\tau_{life}}$$

    Where $\dot{\mathbf{m}}$ is the is the **annual production volume** and τ **is the lifetime of the factory** = multiplied equals to the total production over the lifetime of the factory (either in the operation and demolition).
    - The representation of these coefficients is included in the Aff matrix, diving it in 9 sub matrices (oper. x const. x dem.)

- **Allocation** is the problem that occurs when a process generates more than one product and it is necessary to distinguish the impacts generated by each product. There are several methods to deal with the allocation problem:
    - **Substitution approach ➔** based on calculating the impacts associated with producing the same product using an **alternative technology**. Therefore, we can calculate the contribution of each product subtracting to the total impacts the avoided production of each other processes, using this alternative technology.
    For this reason, it is important to have a solid **rational choice** of the alternative production method applied as well as to perform a sensitivity analysis with respect various alternatives analysis.

$$u_1 = \left(W_{tot} - \sum_{i \neq 1}^{n} u_i^* y_i \right) y_1^{-1}$$

    Thus, we can calculate the impacts by:

$$\beta_{el} = \alpha\beta_{chp}\{y\}_{chp}\{y\}_{el}^{-1} \qquad \beta_{heat} = (1-\alpha)\beta_{chp}\{y\}_{chp}\{y\}_{heat}^{-1}$$

- **Partitioning approach ➔** based on **assigning a share** of the **total** impacts using a certain chosen property like energy or price. This assigning process can be done with the **partitioning variable** which is between 0 and 1. This partitioning can be done **based on certain data**, such as the economic costs or the amount of production generated / consumed by each process (i.e, energy, etc).
  By simply multiplying this coefficient to the total amount of impacts generated and then dividing per the demand of the process, it can be obtained the impacts per unit of demand.

$$u_i^* \cdot y_i = \alpha_i \cdot u_{CHP} \cdot y_{CHP}$$

$$u_i^* = \frac{\alpha_i \cdot u_{CHP} \cdot y_{CHP}}{y_i} \qquad \alpha_i = \frac{y_i \cdot c_i}{y_i \cdot c_i + y_j \cdot c_j} = \frac{y_i \cdot c_i}{y_{CHP} \cdot c_{CHP}}$$

  Finally, we can calculate the impacts by calculating:

$$\beta_{el} = CSL(y_{chp} - y_{heat})\{y\}_{el}^{-1}$$

- **Disaggregation approach ➔** based on retrieving **more detailed inventory** model that allows us to distinguish between processes producing various products. **Separate inventory models** can be established for the two products, and the allocation problem is solved.

- Following the guidelines of ISO 14044:2006, the best procedure for allocation is to avoid it as far as it is possible:
  - 1. By getting **more information** about the internal processes in order to avoid it.
  - 2. If not, the main process has to be **partitioned** in a certain way that reflects the **physical relationship** between them in the system.
  - 3. Finally, if that is also not possible, a **partitioning approach** might be used based in other relationships between inputs and the products.

4. Integrated hybrid LCA

- This method uses a process based foreground system which is directly linked to the background economy.
- This method also allows for modeling systems where the **foreground** system is such **magnitude** that it is not negligible with respect to the background economy.
- Backside of the method: **intensive** and **more complicated** to use than the IO based and tiered hybrid LCA, mainly due to the **compilation of data** needed.
- The background part in the classic LCA analysis is replaced with an input - output data set. The matrix A is then defined as follows:

$$A = \begin{bmatrix} A_{ff} & A_{fn} \\ A_{nf} & \tilde{A}_{nn} \end{bmatrix}$$

-**Aff** = Same meaning as before: interdependency of the foreground processes

-**Afn** = flows from the foreground processes to the background economy.

-**Anf** = contains the inputs from the background economy to the foreground processes.

-**Ann** = inter-sectoral requirements in the background economy.

- The sub matrix **Afn is not zero** because the volume off the foreground system (Aff) is not negligible compared to the flows of the national economy.
- The challenge is to **remove the process** we model because they are also represented in the Ann matrix = remove the processes of foreground system out of the background economy, proportionally to the volume in activity.
    - If we don't do that, the total output x for a given functional unit will be too large, because we will **double counting** the same processes.
- Objective = derive an adjusted flow matrix which represents the whole hybrid system, Z. Thus, we need to introduce additional matrices:
    - Z = complete flow matrix
    - Hfb = process -p to commodities -n correspondence matrix
    - π f = prices of processes in the foreground processes.

$$\tilde{Z}_{bb} = Z_{bb} - \tilde{Z}_{ff} - \tilde{Z}_{fb} - \tilde{Z}_{bf}$$

where

$$\tilde{Z}_{ff} = H'_{fb}\hat{\pi}_f Z_{ff} H_{fb}$$
$$\tilde{Z}_{fb} = H'_{fb}\hat{\pi}_f Z_{fb}$$
$$\tilde{Z}_{bf} = Z_{bf} H_{fb}$$

$$\tilde{Z} = A\tilde{x} \Leftrightarrow A = \tilde{Z}\hat{\tilde{x}}^{-1}$$

$$\tilde{Z} = \begin{bmatrix} Z_{ff} & Z_{fb} \\ Z_{bf} & \tilde{Z}_{bb} \end{bmatrix}$$

- Thus, we can now calculate the Zbb^, the new output vector x^ (also with ^), and the hybrid A coefficient matrix:

$$\tilde{x}_b = x_b - H'_{fb}\hat{\pi}_f x_f \qquad \tilde{x} = \begin{bmatrix} x_f \\ \tilde{x}_b \end{bmatrix} \qquad A = \begin{bmatrix} A_{ff} & A_{fn} \\ A_{nf} & \tilde{A}_{nn} \end{bmatrix}$$